

EdgeAI: Diabetic Retinopathy Detection in Intel Architecture

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Abstract—Diabetic retinopathy is a leading cause of blindness among working-age adults. Millions of people suffer from Diabetic Retinopathy in India. More dreaded situation is faced by the population in rural India where access to quality healthcare is limited. AI comes to the rescue in those situations where initial diagnosis can be performed without much manual intervention. Early detection of this condition is critical for good prognosis. In this paper we propose a solution using UP2 board (Edge device based on x86 architecture) where AI diagnosis can be performed on the local premise itself. We used PyTorch framework for training the model using EfficientNet-B4 network architecture. Trained model was optimized using Intel Distribution of OpenVINO, and hence there is no much compromise in execution time. Inference time for execution of single image in UP2 board is 0.2 sec. Our model achieved test metric performance comparable to baseline literature results, with sensitivity of 91.5% and specificity of 97.86%. For proof of concept we used open dataset from Kaggle 2019 competition hosted by Aravind Hospital, India.

Index Terms—Diabetic Retinopathy, PyTorch, Intel Distribution of OpenVINO, Edge Computing, UP2 board

I. INTRODUCTION

Following the Political Declaration on Noncommunicable Diseases (NCDs) adopted by the UN General Assembly in 2011, WHO developed a global monitoring framework to enable global tracking of progress in preventing and controlling major noncommunicable diseases - cardiovascular disease, cancer, chronic lung diseases and diabetes - and their key risk factors. Diabetic Retinopathy (DR) is the most common microvascular complication in diabetes [1]. Retinal imaging is the most widely used method of retinopathy screening due to its high sensitivity. The evaluation of the severity and degree of retinopathy associated with a person having diabetes, is currently performed by medical experts based on analysing fundus or retinal images of patient's eyes. About one in six people with Diabetes in the world is from India, setting India's rank as second, in the countries among people with Diabetes. An estimate of 77 million people are diagnosed as having diabetes in India. In India, there are an estimated 9,000 to 10,000 ophthalmologists, which forms ratio of one ophthalmologist to 100,000 population. This is well below the World Health Organizations recommended ratio of 1 ophthalmologist per 20,000 population. This shows the serious shortage of trained medical practitioners. Lack of ophthalmologists makes diagnosis difficult in rural areas of India. Also, detecting retinopathy is a time consuming process and is difficult even

for an ophthalmologist to evaluate and examine digital color fundus photographs of retina.

Telemedicine [2] is a potential cost-effective solution to access problems. Patients can have retinal images taken at diabetology center or primary care clinics, and the cases can reviewed by a remote expert. Today, a major impediment in implementing telescreening in tertiary care centers is the lack of trained graders to grade the fundus photography images sent from the remote clinics. Thus, the requirement arises for an automated system to assess the severity of DR which helps to scale retinopathy screenings.

In this current work, we propose a solution to diagnose Diabetic Retinopathy in rural areas using UP squared (UP2) board, leveraging Intel Distribution of Open Visual Inference and Neural Network Optimization (OpenVINO) Toolkit [3].

Rest of the paper has been organized as follows. Section II contains literature survey and Section III describes challenge faced, model training and deployment details have been described in Section IV followed by Results and Conclusion in Section V and VI respectively.

II. RELATED WORK

From researches it is seen that for screening moderate or worse diabetic retinopathy, automated diabetic retinopathy system's performance was equal to or exceeded manual grading [4]. IDx-DR [19], is the first FDA approved autonomous AI system which uses cloud-based software for detecting vision loss. Their system analyzes images from a retinal camera for evidence of lesions. Similarly, the study in [5] represents a critical milestone in using this technology in clinical settings in India. AI algorithms developed for DR screening (for eg, Google AI, EyeArt and IDx-DR) work on cloud-based platforms [6]- [9]. Captured images are uploaded to the cloud and the algorithm provides an output within an acceptable turn over time. But these solutions are difficult in deployment in rural areas since most of the solutions require high end machines / Cloud which is inaccessible in rural areas. Lack of connectivity along with latency and data security issues come to play when using those solutions. This study focuses on training and optimizing deep learning model, and using single board computer like UP2 board for deploying detection model.

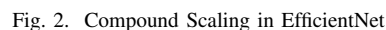
IV. MATERIALS AND METHODS

Fundus photography involves photographing the rear of an eye; also known as the fundus. Specialized fundus cameras consist of an intricate microscope attached to a flash enabled camera. The main structures that can be visualized on a fundus photo are the central and peripheral retina, optic disc and macula. With the advancement of technology, portable fundus camera and smartphone based cameras are available as imaging devices.

We used data hosted by Aravind Hospital, from Kaggle 2019 challenge, for training our Deep Learning model. Since data available with Kaggle 2019 competition [10] was small in number, we used data from Kaggle 2015 Competition [16] for learning the various features better. Kaggle dataset access, data use and security are subjected to the rules mentioned in [17]. Fundus image Dataset consist of color photographs that vary in height and width between low hundreds to low thousands. Kaggle dataset consists of a larger proportion of uninterpretable images due to artifact preponderance, faulty labeling and poor quality [11]. Total of 2930 images from 2019 APTOS dataset and 35126 images from EyePACS dataset was used for training. Clinician rated each image on the basis of severity of diabetic retinopathy according to the 5-point ICDR grade. The granular 5-point grading would be helpful for screening programs in which patient treatment varies at each level of severity. In particular, the threshold for and timing of referrals in DR screening programs often depend on the resources of the program. Currently, in most programs, sight-threatening cases will be referred urgently. Identifying mild cases may also be of clinical value for health care systems that have a different screening interval for patients with no and mild disease. Classes into which the fundus images are graded have been added below.

- Kaggle 2015 competition data was provided by EyePACS, a free platform for retinopathy screening, and 2019 Kaggle Data was collected from Aravind Hospital, Chennai.

Mingxing Tan and Quoc V. Le of Google Research, Brain team designed a Baseline network by performing the neural architecture search, a technique for automating the design of neural networks. This optimized both the accuracy and efficiency as measured on the floating-point operations per second (FLOPS) basis. This architecture uses the mobile inverted bottleneck convolution (MBConv). The researchers then scaled up baseline network to obtain a family of deep learning models, called EfficientNet [12]. EfficientNet is a type of CNN architecture. It's architecture is given in Fig. 1. Here compound scaling method as shown in Fig. 2 is used. Compound scaling method improved model's accuracy and efficiency. EfficientNet has more control over model performance by having constraints on the scaling coefficients which involves weighted scale of three inter-connected hyper parameters of the model - Resolution, Depth and Width of the Network. Unlike other network architecture, EfficientNet has lowest number of training parameters with high accuracy. Adding to this, EfficientNet has the highest top-1 accuracy in ImageNet data. This prompted to choose EfficientNet as pre-trained DL model for retinopathy classification.



UP2 board (Single board computer with high performance and low power consumption) with Intel Pentium Processor is an edge device from Aaeon Technology Inc, in x86 architecture. UP Squared (UP2) has high performance and low power consumption features of Intel Celeron, Pentium and Atom Processors (codename Apollo Lake). This device is available in 8GB LPDDR4 configuration. UP Squared supports AI accelerator devices such as AI Core X mPCIe module.

which features Intel Movidius Myriad X. With speeds up to 105 fps (80 typical) and 1 trillion floating point operations, adding on the AI Core X to UP Squared creates a powerful, compact AI Edge solution.

E. Intel Distribution of OpenVINO

Intel Distribution of OpenVINO toolkit [3], [13], [14], [15] is an open-source product from Intel which contains Deep Learning Deployment Toolkit (DLDT) for Intel processors (for CPUs), Intel Processor Graphics (for GPUs), and heterogeneous support. It includes an open model zoo with pre-trained models, samples, and demos. This toolkit allows developers to deploy pre-trained deep learning models through a high-level C++ or Python inference engine API integrated with application logic. Intel Distribution of OpenVINO employs model optimizer and Inference engine for running Inference code. HW pre-requisite for installation is that Intel Distribution of OpenVINO requires 6th generation or above Intel Core processors.

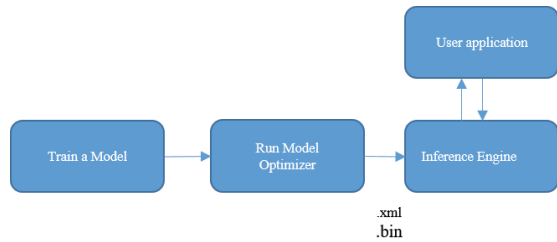


Fig. 3. Workflow for deploying Deep Learning Model in Intel Distribution of OpenVINO

As shown in Fig. 3, Model Optimizer (MO) converts trained model from a specific framework (like Caffe or TensorFlow) to a framework-agnostic Intermediate Representation (IR) format which is used as an input for the Inference Engine. Model Optimizer generates two files: xml and bin file. xml is the topology file that describes the network topology and .bin is the trained data file which consist of weights and biases. MO performs a number of device-agnostic optimizations. For example, certain primitives like linear operations (BatchNorm and ScaleShift), are automatically fused into convolutions. Generally, these layers should not be manifested in the resulting IR. After conversion, Inference Engine consumes IR to perform inference. While Inference Engine API itself is target-agnostic, internally, it has a notion of plugins, which are device-specific libraries(for CPU, GPU, Intel Movidius Myriad2 VPU, Intel Movidius Myriad X VPU and FPGA) facilitating hardware-assisted acceleration. Intel Distribution of OpenVINO toolkit includes the Intel MKL-DNN, a high-performance library designed to accelerate neural network primitives, increase application performance, and reduce development time.

Intel Distribution of OpenVINO toolkit for Linux [3], [13]–[15]:

- Enables CNN-based deep learning inference on the edge

- Supports heterogeneous execution across Intel CPU, Intel Integrated Graphics, Intel Movidius Neural Compute Stick, Intel Neural Compute Stick 2, and Intel Vision Accelerator Design with Intel Movidius VPUs
- Speeds time-to-market via an easy-to-use library of computer vision functions and pre-optimized kernels
- Includes optimized calls for computer vision standards including OpenCV*, OpenCL, and OpenVX

F. Workflow for proposed system

Fig. 4 shows the workflow diagram of our proposed work. Images are preprocessed and fed as input to the DL model. DL model learns the features directly from the data without the need for manual feature extraction. Deep learning algorithms seek to exploit the unknown structure in the input distribution, in order to discover good representations, often at multiple levels, with higher-level learned features defined in terms of lower-level features [18]. Once the model is trained and validated, it is converted to IR model. As shown in Fig. 4, during runtime, input fundus images are fed to EfficientNet-B4 IR optimized model for leveraging optimized model performance. Below subsections gives detailed information on each of the processes involved in the workflow.

G. Preprocessing

Table I and Table II shows the image count taken in Train / Val / Test cases. Preprocessing consisted of finding an approximate radius of the retina and scaling down the image. This involved finding the centroid of the fundus region and extracting tightly contained circular area of fundus. Since all the images were of different size, we did rescaling of images to the same size i.e. 288x288. Different augmentation techniques were applied to augment the data for creating different input variations. We applied random affine transformation comprising of rotation, scaling, shearing, Horizontal and Vertical Flip along with adding Color Jitter. Patch of size 256x256 was extracted from the center, normalized and fed as input to deep learning model for training.

TABLE I
KAGGLE 2015 CHALLENGE IMAGE COUNT

Category	Normal	Mild	Moderate	Severe	Proliferative
Train	25810	2443	5292	873	708

TABLE II
KAGGLE 2019 CHALLENGE IMAGE COUNT

Category	Normal	Mild	Moderate	Severe	Proliferative
Train	1166	235	630	122	190
Val	292	59	158	31	47
Test	347	76	211	40	58

H. Model Training

We used PyTorch framework to train the model. Model was trained using Kaggle 2015 and 2019 dataset. Table IV below

show hyperparameter configuration. EfficientNet-B4 architecture was used for model training with MSE loss function. Diabetic Retinopathy problem has significance in ordering of labels i.e. Severe DR is more severe than moderate DR, but less severe than Proliferative DR. Since in classification problem, ordering of the labels is thrown away, we treated the problem as regression problem and used pre-defined threshold to capture the class id. Activation layer of EfficientNet-B4 is swish activation. Output of this activation layer is thresholded to obtain the result. Following threshold values (0.5, 1.5, 2.5, 3.5) was used to discretize the predicted regression values and make the class level into integer. Optimizer used was Rectified ADAM. Rectified Adam (RAdam), is a variant of the Adam optimizer, as it rectifies the variance/generalization issues apparent in other adaptive learning rate optimizers. We completed training in eight levels of 30 epochs each. The training loss didn't lower further. We used cosine annealing scheduler for learning rate updation. Cosine annealing is an example of aggressive learning rate schedule where learning rate starts high and is dropped relatively rapidly to a minimum value near zero before being increased again to the maximum.

TABLE III

Architecture	x86_64
Model name	Intel(R) Pentium(R) CPU N4200 @ 1.10GHz
Core/socket	4
Thread/socket	4
Intel Distribution of OpenVINO	2019 R3

TABLE IV
HYPERPARAMETER CONFIGURATION

Hyperparameter	Configuration parameter
Architecture	EfficientNet-B4
Loss function	MSE
Optimizer	RAdam
Learning Rate Decay	Cosine Annealing LR
Batch Size	32
Epoch	240

Cross Validation : Inorder to assess the model, 5 fold cross validation was done on 2019 data. Validation accuracy for 2019 data after 5 fold cross validation is 92.5%.

I. Model Performance

Table. V shows performance of the trained model.

1) *Quadratic Weighted Kappa*: Quadratic weighted kappa, the state-of-the-art performance metric for multi-class classification and suggested evaluation metric for Diabetic Retinopathy Kaggle problem was adopted as the performance metric of severity grading prediction. Quadratic Weighted Kappa, measures the agreement between two ratings. The metric typically varies from 0 (random agreement between raters) to 1 (complete agreement between raters). In the event that there

is less agreement between the raters than expected by chance, the metric may go below 0. The quadratic weighted kappa is calculated between the scores which are expected/known and the predicted scores.

J. QuEST Solution Accelerated by Intel Distribution of OpenVINO Toolkit

Taking cue from OpenVINO toolkit we accelerated inference of our DL model for Diabetic Retinopathy Classification in UP2 board leveraging Intel Distribution OpenVINO, to employ model inferencing in low end Intel machines without much compromise in performance. Figure 4 shows the workflow diagram of our solution. As shown in the Figure, training was done offline in NVIDIA GTX 1080Ti.

Once the model is trained, EfficientNet-B4 PyTorch model is converted to ONNX model. The ONNX model is then converted to OpenVINO IR model using model optimizer. Once the model is converted to IR format this can be read using Inference Engine of OpenVINO.

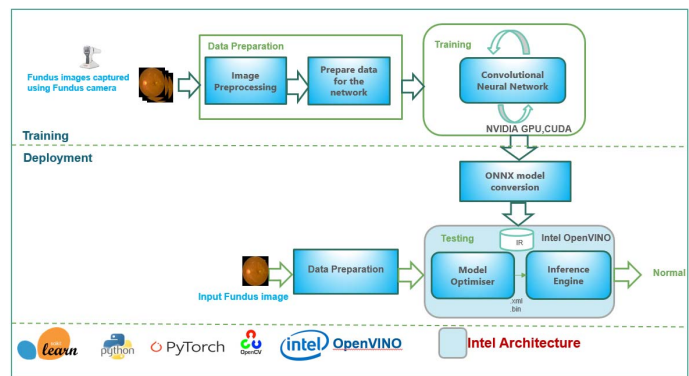


Fig. 4. Work flow Diagram of AI application

K. Rural Area Deployment

The use case has been depicted in the Figure 5. Now portable fundus-on-phone (FOP)/smartphones are available for capturing Fundus images along with tabletop fundus camera. These cameras can be used during rural area eye camps for patient screening. Fundus image captured using fundus cameras / smartphone cameras can be connected to the AI powered edge device. Edge device would contain the model pre-trained and optimized for deployment. Hence during inference, on inputting an image, our model would identify the class to which the fundus image belongs. Our application provides an easy interface for anyone to analyze retinal scans, and to make the DR screening process faster and efficient. And based on the inference, those patients can be referred to the hospital wherein further diagnosis and treatment can be prescribed by the ophthalmologist.

V. RESULTS

Test images set apart from Kaggle 2019 data is fed as input to Intel Distribution of OpenVINO optimized IR model to run on UP2 board(Intel Pentium N4200/5). Total of 732

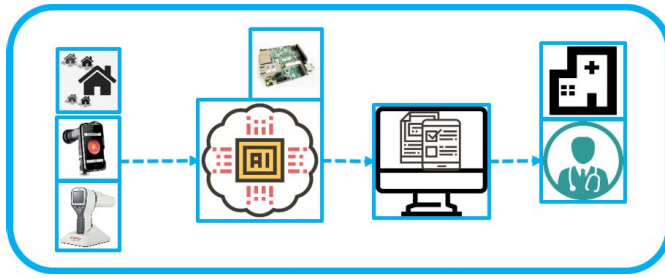


Fig. 5. Rural Deployment of AI model

images was used for testing and 537 images were set apart for Validation. Model was trained using PyTorch framework in NVIDIA GTX 1080Ti. 5 fold Cross Validation accuracy with Kaggle 2019 data was 92.5%. Once the model has been validated, PyTorch model was converted to ONNX model and then to OpenVINO optimized IR model. This IR model is deployed in UP2 board. The preprocessed images are fed to pre-trained OpenVINO optimised EfficientNet-B4 model for inference. Execution in UP2 took about 0.2 s. High sensitivity and specificity of DL model are essential for effective retinopathy screening. Our DR model gave sensitivity of 91.5% [95% CI, 89.27% to 93.44%] and specificity of 97.8% [95% CI, 97.29% to 98.37%]. Fig. 6 shows the ROC curve for EfficientNet-B4 model. Area Under the Receiver Operating Characteristic Curve (AUC) is 0.95. AUC metric provides an aggregate measure of performance across all possible classification thresholds. Quadratic Kappa score of the model, measured comes to 0.96. This indicates that our model prediction is almost in agreement with groundtruth for most of the cases. Table III contains machine configuration along with version details of Framework / Topology / Tools / Library used for inference. Table V and VI contains performance metric of our model. VII compares the execution time of the model in Core i7 machine and UP2 board. It is noted that with performance-cost-portability tradeoff user can deploy UP2 board in rural area eye camps. This also shows that performance of the model improves when processor with better hardware spec is used. User can select the HW considering the following factors i.e. portability, cost factor and compact factor for AI deployment.

TABLE V
MODEL PERFORMANCE

Model	Validation Accuracy 2019 Data	Test Accuracy 2019 Data	Quadratic Weighted Kappa score (2019 Test Data)
EfficientNet-B4	92.5%	91.5%	0.96

VI. CONCLUSION AND FUTURE WORK

With the state of the art deep learning technology we have worked on Diabetic Retinopathy Detection problem to come

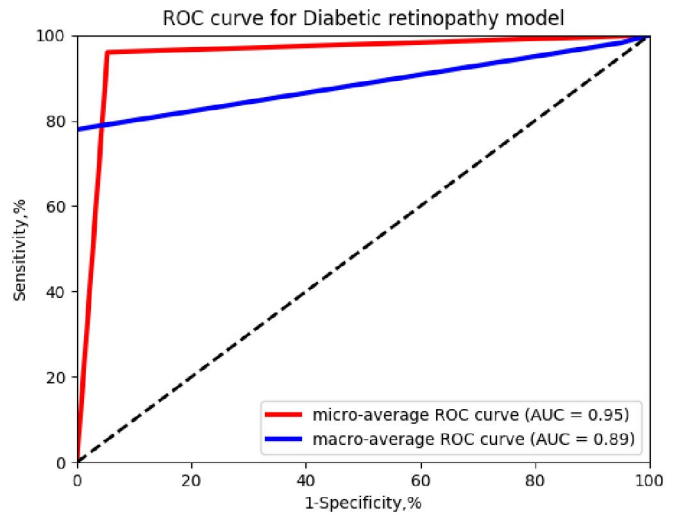


Fig. 6. ROC Curve of EfficientNet-B4 trained model

TABLE VI
PERFORMANCE METRIC

Quadratic Kappa score	0.96
Sensitivity	91.5%
Specificity	97.86%
Precision score	92%

up with an AI solution on edge. Model has been trained using EfficientNet-B4 network architecture. Deployment on edge device has been done after model optimization using Intel Distribution of OpenVINO. This improves execution time on the device and provides portable AI solution during rural area eye camps. Current scope of our work has been proof of concept for edge AI deployment. In future, we can do re-training of our model using new set of images captured from portable fundus camera and integrate it with the hardware unit for retinopathy screening.

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TABLE VII
EXECUTION TIME IN DIFFERENT MACHINE CONFIGURATIONS

Machine Details	Execution Time (sec)
Optimization on Intel Core i7 with Intel Distribution of OpenVINO	0.03
Optimization on UP2 with Intel Distribution of OpenVINO	0.2

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